Nahee **Kim**

M.S. Student in Data Science · Trustworthy AI & State-Space Models

Incheon, Republic of Korea

knh1980@inha.edu | i-nae.github.io | Personal Blog

"I study internal model dynamics through interpretable signals and physics-inspired principles, with the goal of detecting behavioral shifts and enabling more reliable and controllable AI systems."

Research Profile

I study how internal model signals reflect changes in neural behavior, focusing on interpretability, reliability, and robustness. My research develops physically and geometrically grounded methods to identify failure-prone inputs under noise and distribution shifts. I aim to extend these white-box methods to foundation models and AI agents for behavioral monitoring, safety evaluation, and control.

Education

Inha University — *Incheon, Republic of Korea*

M.S. in Data Science — *Sep. 2025 – Feb. 2027 (Expected)*

- Expected to complete the M.S. degree in three semesters through early graduation
- Research focus: trustworthy AI, state-space models, model interpretability, and scientific imaging
- Advisor: Prof. Ji Hyun Nam
- GPA: 4.13 / 4.5

Inha University — *Incheon, Republic of Korea*

B.S. in Electronic Engineering — *Mar. 2019 – Aug. 2025*

- GPA: 3.80 / 4.5

Research Experience

RiemannMamba: Geometry-Aware Internal Time in Selective State Space Models — *Submitted to NeurIPS 2026*

Sole First Author — *2025 – Present*

- **Architectural reinterpretation:** Reinterpreted Mamba's input-dependent step size Δ not merely as a selective gate, but as a state-dependent geometry on the hidden-state space.
- **Reliability:** Accumulated geometry-weighted hidden-state changes into an internal time variable τ, aligned it with sample difficulty and error propensity, and applied it to selective prediction under distribution shifts and corruptions.
- **Interpretability:** Developed gradient-free τ-CAM to spatially visualize the internal time signal and identify image regions that induce stronger hidden-state evolution.

TST-Mamba: Plug-and-Play Selective State-Space Denoising for Non-Line-of-Sight Imaging — *Submitted to Neurocomputing*

Sole First Author — *2025 – Present*

- **NLOS noise research:** Characterized the dominant noise patterns in real NLOS measurements and developed a real-measurement-guided SPAD noise simulator tailored to transient denoising.
- **Architecture:** Designed and formalized TST-Mamba as a Mamba-based skip-connected denoising autoencoder for NLOS transients, combining an undercomplete temporal bottleneck, single-pass spatial processing in latent space, and skip–latent fusion to preserve reconstruction-critical temporal details.
- **Cross-pipeline robustness and generalization:** Demonstrated that TST-Mamba operates as a reconstruction-agnostic plug-and-play denoiser, improving downstream reconstruction without modifying or retraining each pipeline and maintaining effectiveness across simulated, unseen-category, and real NLOS measurements.

Driver Handover Readiness Estimation via Bio–Vehicle Signal Fusion and Safety–Oriented Re-decision — *IEIE Summer Annual Conference 2026*

Lead Researcher · Accepted Conference Paper — *Jun. 2026*

- **Project leadership:** Conceived and led the project end-to-end, including problem formulation, framework design, implementation, experimental evaluation, result analysis, and manuscript writing.
- **Methodology:** Developed a bio–vehicle signal fusion framework combining ECG-derived driver stress estimates with vehicle-state information, and designed a reward-based re-decision policy to reduce unsafe handover approvals.
- **Validation and presentation:** Evaluated the framework through driver-held-out experiments using WESAD and OpenDriver data; selected for the Best Paper Award Presentation Session (VBP), Jeju, Republic of Korea.

Air Purifier Utilizing Tesla Compressor and Cyclone Dust Collector — *Technology and Education · KSME*

KCI Journal Paper — *2024*

- **Research contribution:** Implemented motor control for the Tesla compressor and independently conducted the experimental analysis and manuscript writing.

Additional Research Experience

SocialWise: Augmented-Reality-Based Social Communication Support for Children with Autism Spectrum Disorder *Inha University*

Undergraduate Research Assistant *2024*

- **Human-centered assistive AI:** Contributed to the development of an augmented-reality-based system designed to interpret social interaction cues and support social communication for children with autism spectrum disorder.
- **Speech data collection:** Conducted speech-utterance data collection with participants with autism spectrum disorder for the development of the assistive communication system.

Publications & Manuscripts

[1] **Nahee Kim** and Ji Hyun Nam. "RiemannMamba: Geometry-Aware Internal Time in Selective State Space Models." Manuscript submitted to *NeurIPS 2026*.

[2] **Nahee Kim** and Ji Hyun Nam. "TST-Mamba: Plug-and-Play Selective State-Space Denoising for Non-Line-of-Sight Imaging." Manuscript submitted to *Neurocomputing*.

[3] WooSuk Choi, **Nahee Kim**, and Ji Hyun Nam. "Driver Handover Readiness Estimation via Bio–Vehicle Signal Fusion and Safety–Oriented Re-decision." Accepted to the *IEIE Summer Annual Conference 2026*.

[4] Je-Yoon Ryu, Hyunu Kim, Hyun-Wook Chung, **Nahee Kim**, Jae-Hyeon An, and Sun-Kon Lee. "Air Purifier Utilizing Tesla Compressor and Cyclone Dust Collector." *Technology and Education*, vol. 12, no. 2, pp. 63–72, 2024. KCI-indexed.

Teaching Experience

Object-Oriented Programming *Inha University*

Teaching Assistant *Spring 2026*

- Assisted students with object-oriented programming concepts, programming exercises, and the implementation and debugging of course assignments.

Image Data Analysis *Inha University*

Teaching Assistant *Spring 2026*

- Supported coursework on image-data processing and analysis, helping students understand computational methods, experimental workflows, and the interpretation of image-based results.

Physics Laboratory *Inha University*

Teaching Assistant *Fall 2025*

- Led full undergraduate physics laboratory sessions, including explaining experimental principles and procedures, supervising measurements and equipment use, guiding data analysis, and evaluating students' experimental work.

High School Physics *Mirae Injae Academy*

Physics Instructor *May 2023 – Apr. 2024*

- Taught high school students fundamental physics concepts and problem-solving methods, explaining mathematical and physical principles in an accessible and structured manner.

Honors & Awards

Oct. 2023	**Bronze Award**, 13th National Student Design Competition (Tesla-Compressor Cyclone Collector)	*KSME*
Oct. 2023	**Excellence Award**, 15th Creative Design Competition (Multi-Phase Hybrid Generator)	*I-DREAM*
Sep. 2020	**Bronze Award**, 2020 Capstone Design Competition (Automatic Nozzle Changer for 3D Printers)	*Inha University*
Jul. 2020	**Grand Prize**, 2020 Engineering Design Club Competition (Automatic Nozzle Changer for 3D Printers)	*Inha Univ. ICEE*

Skills & Research Expertise

Programming	Python, C, C++
Deep Learning	PyTorch, selective state-space models, Mamba architectures, CNNs, and spatio-temporal modeling
Trustworthy AI	Interpretability, selective prediction, reliability estimation, and robustness under corruptions
Scientific Imaging	NLOS imaging, transient-data processing, SPAD noise simulation, reconstruction, and denoising
Languages	Korean (native), English (working proficiency)